

Chapter 1: The Secret of Computer Viruses

An Interactive Smart Board Class on staying safe in our digital world. Adapted directly from Chapter 1 content.

Today's Learning Goals

✂ Animation: Fly In (Each Card)



Understand

Discover what a computer virus is and why it behaves like a biological cold.



Identify

Learn to spot different sneaky malware files like Worms, Trojans, and Spyware.



Defend

Master smart safety rules and use Antivirus software to lock out threats.

TEACHER GUIDE

Ask students if they have ever had a common cold. Bridge that biological cold experience into how a computer might catch a "digital cold."



Warm-up: Can Computers Get Sick?

✂ Animation: Fade In

Let's discuss three quick questions:

- ✓ Have you ever caught a cold? How did it feel? Did it slow you down?
- ✓ How do colds pass from one friend to another?
- ✓ If a biological virus makes us sneeze, what does a computer virus do?

💡 Daily Life Connection

Passing a cold by sharing a cup is just like passing a computer virus by sharing a contaminated pen drive!



Biological vs. Digital

Both replicate silently, spread easily, and need special defense systems to keep healthy!

INTERACTIVE PROMPT

Tap on the Smart Board to reveal the comparison. Encourage students to call out answers before clicking.

What is a Computer Virus?

✂ Animation: Zoom (Image) + Fade (Text)

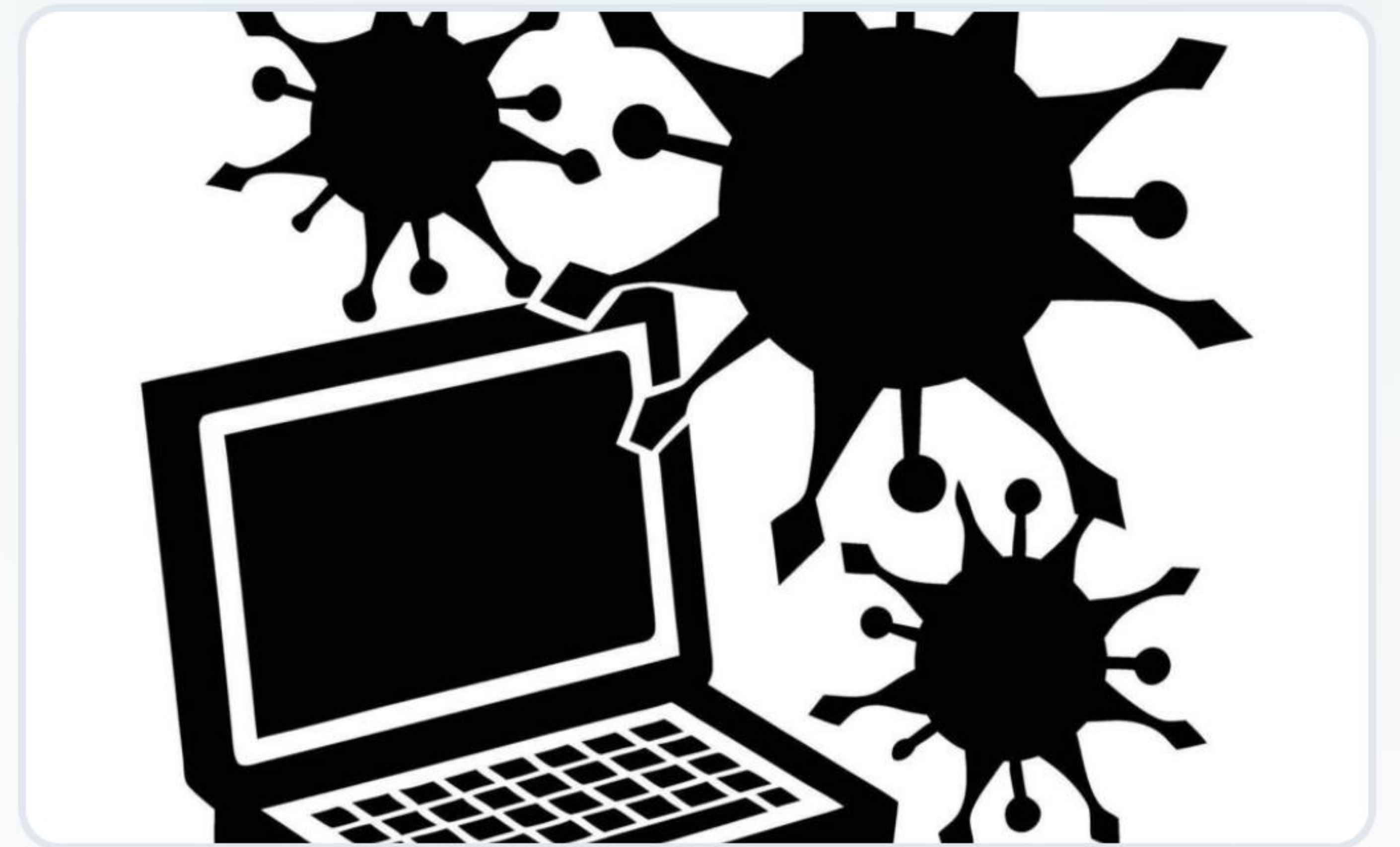
A **Computer Virus** is a tiny, hidden software program written on purpose to corrupt files and steal resources.

The Full Form of V.I.R.U.S.

Vital Information Resources Under Siege

This means your precious files (information) are being surrounded and attacked by sneaky code!

- ✓ **It is Executable:** It runs automatically when attached to software.
- ✓ **It's Malicious:** Designed to erase data, move files, and eat memory.



LESSON CONTEXT

Refer to Page 7 of *WhatsApp Image 2026-06-22 at 9.34.08 AM.jpeg* . Remind them that a computer virus cannot exist without human-written code.



How Do Viruses Spread?

✂ Animation: Appear one-by-one



Pen Drives

Transferring files from infected flash drives or external storage is a super common way to pass viruses.



Spam Emails

The most common way! Opening weird-looking attachments from strangers instantly activates bad code.



Pirated Sites

Downloading illegal "cracked" games, movie files, or untrusted browser plugins drops viruses on your disk.

⚠ Classroom Discussion Question

If you find a random pen drive on the playground, is it safe to plug it into your computer? Why or why not?

Quick Quiz: Are You Ready?

🔗 Interactive Smart Board Mode

Question 1: What does the letters 'V.I.R.U.S' stand for?

A

Very Important
Resource

B

Vital Information
Resources Under Siege

True or False: A computer virus is created naturally by dust inside the PC.
(Tap answer on board to discuss)



Points Challenge

Let's see which row gets the first correct match! Ask students to explain *why* B is correct.

ANSWERS

Q1 is B. True/False is False (it's custom software written intentionally by programmers). Reference: Page 12 of

WhatsApp Image 2026-06-22 at 9.34.08 AM (1).jpeg



Meet the **Malware Family**

✂ Animation: Fly In

Malware (Malicious Software) is the family name for all harmful computer programs. Let's see who is in this sneaky family:



Worms

Unlike viruses, they don't need to attach to a host. They copy themselves and crawl across networks quickly.



Trojan Horse

Pretends to be a cool game or file, but once run, it releases hidden code to attack your PC. Like a trap-gift!



Logic Bombs

Secret codes that stay silent until a specific trigger happens (like a calendar date) before exploding to erase data.

REAL-LIFE STORY

Explain the ancient Greek Trojan Horse story (soldiers hiding inside a giant wooden horse) so students remember the metaphor!



Spyware

&

Secret Agents

✂ Layout: Bleed Image Right

Spyware is like a secret digital agent hiding in your computer. It silently records everything you type, including your games, passwords, and private messages!

Did You Know?

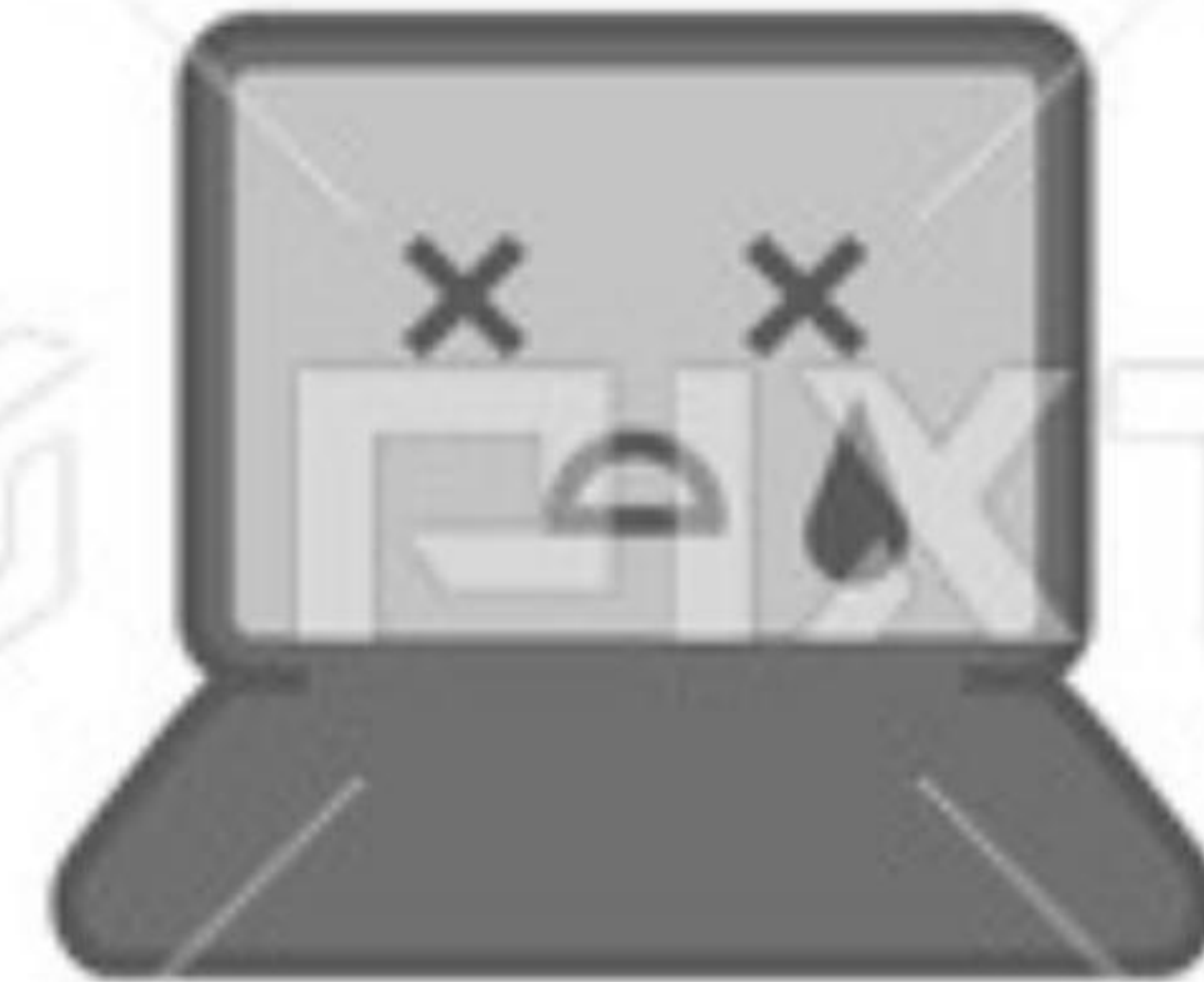
Adware is a type of spyware that displays irritating pop-up ads all over your screen while you're trying to study!

Referenced from Page 9 of *WhatsApp Image 2026-06-22 at 9.34.07 AM.jpeg*.

virus



trojan horse



worm



spyware



Virus Types: Quick Comparison

✂ Animation: Fade In Table

Virus Category	Target Area in Computer	Easy Real-Life Analogy
Program Virus (.exe, .com)	Specific software and application files	A bug that infects specific textbook pages.
Boot Sector Virus	Hard disk startup areas (MBR)	Locking the main school gate so no one can enter.
FAT Virus (File Allocation Table)	Your disk index directory map	Erasing the library index card catalog completely.

KEY CONCEPT

Check Page 8 of *WhatsApp Image 2026-06-22 at 9.34.07 AM.jpeg* . Remind students that a boot sector virus prevents the operating system from booting up!

! Is Your Computer Sick?

✂ Animation: Appear from Left

Just like humans run a fever when sick, computers show warning signs if infected:



Slow Performance

The computer takes ages to turn on, load programs, or gets "hanged" frequently.



Unwanted Pop-ups

Random files and warning windows open by themselves without you clicking anything.



Hidden Data

Files and folders start disappearing or become inaccessible to open.

🔒 Safe Zone Fact

Compressed files and physical hardware components (like cables) can NEVER be infected by a virus! (Page 8)

Quiz: Malware & Signs

Question 2: Which malicious program can self-replicate without attaching to other files?

A Worms

B Trojans

Question 3: Can a virus damage your physical keyboard or screen?

A Yes, it can melt them

B No, only software files

ANSWER KEY

Q2 is A (Worms), Q3 is B. Reinforce that hardware remains physically safe from virtual codes. Reference: Page 8-9.



Defend Your Digital World

✂ Animation: Slide In

Follow these simple steps to block out malware attacks:

- ✓ **Turn On Firewall:** A wall that inspects all incoming web traffic.
- ✓ **Never Open Unknown Emails:** Delete attachments from strangers.
- ✓ **Keep Windows Updated:** Updates are patches that repair holes in the security.
- ✓ **Use Pop-up Blockers:** Stop malicious tabs from jumping onto screens.

Real-life Safety Action

Always check with parents or teachers before executing any downloaded .exe setup files!



INTERACTIVE PROMPT

Point out the digital firewall metaphor. It acts like a school gatekeeper check. Refer to Page 9.



Your Shield: **Antivirus**

✂ Animation: Zoom in

Antivirus software is a utility program that acts like a doctor for your files!

It keeps you healthy by performing three roles:

- ✓ **Scanning:** Regularly searches your whole disk for hidden virus codes.
- ✓ **Identifying:** Detects malware lists by comparing them with an index.
- ✓ **Removing:** Safely quarantines or deletes viruses before they cause harm.



Popular Antivirus Brands

⚡ Animation: Appear one-by-one

Here are some common shields used worldwide to keep computers clean and healthy:



McAfee

A global defense shield checking and blockading malicious web files.



Avast

Famous free shield program with fast real-time scanning.



Kaspersky

Strong security protector analyzing behavior and logic threats.



Other brands mentioned on Page 10: **AVG, Norton, Avira, ESET NOD32, Quick Heal**



Quick Summary Map

✂ Animation: Fade In

✓ Key Concepts

- Virus stands for Vital Information Resources Under Siege.
- It spreads via pen drives, dirty downloads, and email attachments.
- Malware encompasses Viruses, Worms, Trojans, and Spyware.
- Firewalls and Antiviruses are your main line of defense.

⚠ Sneaky Attacks

- Worms replicate alone across networks.
- Trojans act like fake gifts to deliver payloads.
- Spyware record logs silently.
- Logic Bombs activate on specific days/dates.

REVISION TOOL

Ask three students to summarize each section in their own words before jumping into the final challenge exercises!



Complete Mind Map



Computer Safety Master Map

⚙️ Threats

Viruses, Worms, Trojans, Spyware, Logic Bombs



Transmission

Pen drives, Email links, Unknown Downloads



Active Defenses

Firewalls, Pop-up Blockers, Regular Updates, Antivirus Brands

🔍 Discussion Point

Why do you think viruses need human actions to spread, while worms can crawl through wires all by themselves?



Mega Quiz: Part 1

🔧 Interactive Smart Board Mode

Q1. Fill in: A _____ copies itself and infects without user knowledge.

A Computer Virus

B Antivirus software

Q2. True/False: Worms can travel on email/networks without help.

A True

B False

ANSWERS

Q1 is Computer Virus, Q2 is True. Directly crosscheck with questions on Page 12 of *WhatsApp Image 2026-06-22 at 9.34.08 AM (1).jpeg*



Mega Quiz: Part 2

Q3. Which of the following is Malicious Software family name?

A Shareware

B Malware

Q4. True/False: Updating existing Antivirus is not necessary.

A False

B True

CONTEXT INFO

Reinforce standard safety guidelines: updates provide immunizations to stay protected.



Practice Exercises

✂ Animation: Fade In Questions

Short Answers

- ✓ Define "Computer Virus". (Reference Page 7)
- ✓ What is an executable program?
- ✓ List three common symptoms of computer virus infection.
- ✓ What is a firewall and how does it safeguard your connection?

HOTS (Higher Order Thinking)

Let's think out-of-the-box:

- ✓ If compressed files cannot be damaged by common viruses, why don't we compress all files?
- ✓ How does pirated/copied game setup files pose a critical security threat to users?

CLASSROOM STRATEGY

Divide the classroom into group A and group B. Have each group tackle one of the HOTS questions together.



Action Plan: Activity & Homework

✂ Animation: Zoom Cards

🛡 Class Activity: Shield Up!

Look at Page 10 Activity! Draw a shield next to the brands you recognize. Talk to your partner about which Antivirus software runs on your school computer lab systems.

🏠 Homework Task

- Check if your home computer has a Firewall turned on.
- Write down the name of your antivirus software (or Windows Defender) in your notebook.
- Solve the match-the-following on Page 12.

STUDENT NOTE

Homework references Page 12 tasks directly from *WhatsApp Image 2026-06-22 at 9.34.08 AM (1).jpeg*

Questions & Answers?

Awesome job today, digital detectives! Raise your hand if you have any questions about keeping your computer safe and clean.

Computer Science Class • Chapter 1 Complete

Image Sources



https://static.vecteezy.com/system/resources/previews/057/747/046/non_2x/intense-computer-virus-attacking-a-digital-system-portrayed-in-a-stark-flat-silhouette-free-vector.jpg

Source: www.vecteezy.com



Thumbnail for <https://t.pimg.jp/115/085/603/1/115085603.jpg>
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